

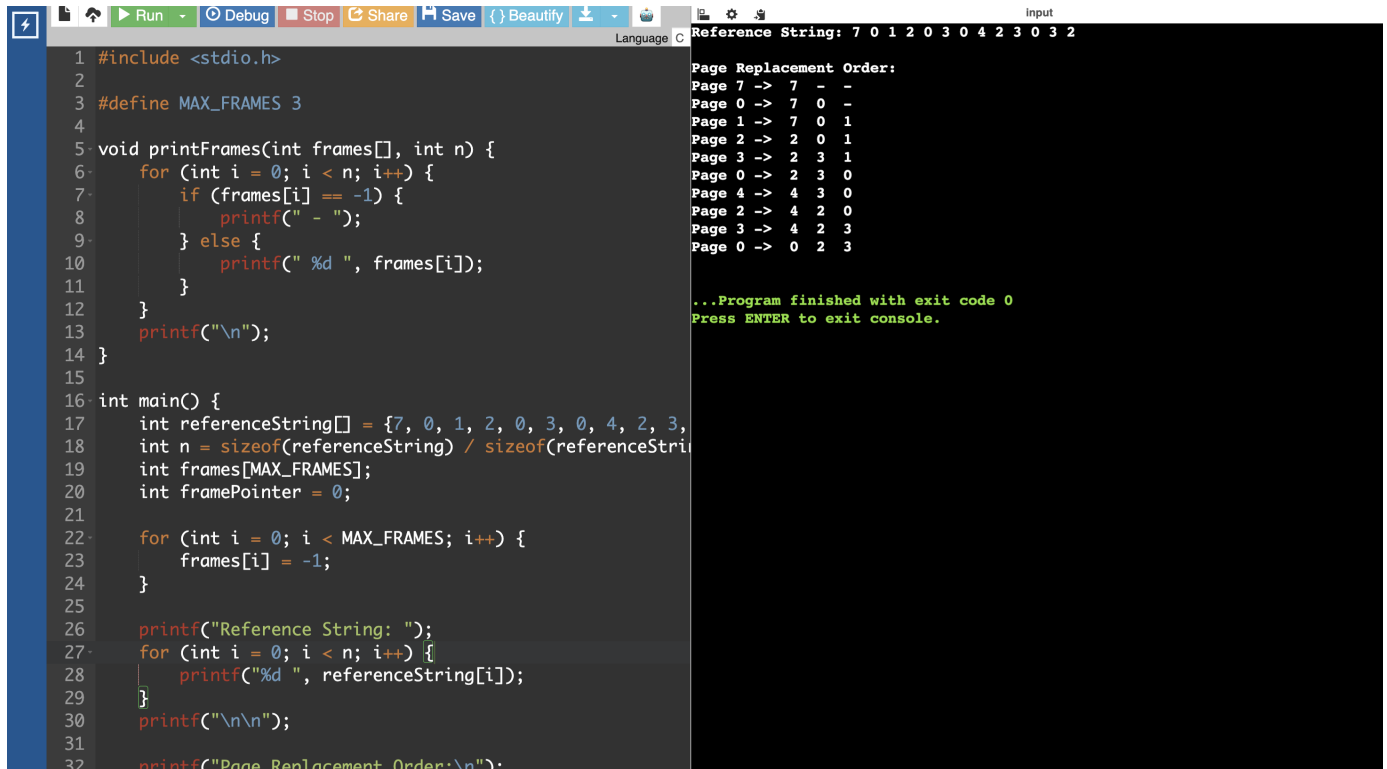
CSA0404 – OPERATING SYSTEMS

EXPERIMENT 31

QUESTION

Construct a C program to simulate the First in First Out paging technique of memory management.

PROGRAM



```
1 #include <stdio.h>
2
3 #define MAX_FRAMES 3
4
5 void printFrames(int frames[], int n) {
6     for (int i = 0; i < n; i++) {
7         if (frames[i] == -1) {
8             printf(" - ");
9         } else {
10            printf(" %d ", frames[i]);
11        }
12    }
13    printf("\n");
14 }
15
16 int main() {
17     int referenceString[] = {7, 0, 1, 2, 0, 3, 0, 4, 2, 3};
18     int n = sizeof(referenceString) / sizeof(referenceString[0]);
19     int frames[MAX_FRAMES];
20     int framePointer = 0;
21
22     for (int i = 0; i < MAX_FRAMES; i++) {
23         frames[i] = -1;
24     }
25
26     printf("Reference String: ");
27     for (int i = 0; i < n; i++) {
28         printf("%d ", referenceString[i]);
29     }
30     printf("\n\n");
31
32     printf("Page Replacement Order:\n");
33
34     // Page Replacement Order:
35     // Page 7 -> 7 - -
36     // Page 0 -> 7 0 -
37     // Page 1 -> 7 0 1
38     // Page 2 -> 2 0 1
39     // Page 3 -> 2 3 1
40     // Page 0 -> 2 3 0
41     // Page 4 -> 4 3 0
42     // Page 2 -> 4 2 0
43     // Page 3 -> 4 2 3
44     // Page 0 -> 0 2 3
45
46     ...Program finished with exit code 0
47     Press ENTER to exit console.
```